

Internship Final Report: ASIC Physical Design (RTL-to-GDSII)

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Contents

Abstract Summary	3
1 Corporate Context	3
2 Department Oversight	3
3 Task Matrices	3
3.1 RTL Datapath & Control Architecture	3
3.2 Synthesis & Foundational Physical Constraints	4
3.3 RISC-V Capstone: Floorplanning & Power Mesh	5
3.4 Routing, DFM, & Physical Signoff	6
4 Curriculum Connection	7
5 Practical Innovation Contribution	7
6 Formal Recognitions	7
7 Concluding Remarks	8
8 References & Sourcing Integrity	9

Abstract Summary

This report details a comprehensive four-week practical internship completed at the NUST Chip Design Centre (NCDC) under the NIPIS '26 program. The training focused on practical Application-Specific Integrated Circuit (ASIC) physical design implementation, bridging the gap between front-end behavioral logic and foundry-ready GDSII layouts. Utilizing the TSMC 65nm LP node and the complete Cadence EDA toolchain, the capstone project involved the physical implementation of a 32-bit RISC-V Single-Cycle Processor [4]. Key objectives included power distribution network (PDN) generation, standard cell placement, zero-skew clock tree synthesis (CTS), and detailed routing using NanoRoute [4]. The workflow culminated in successful multi-corner static timing analysis (STA) and physical signoff [4].

1 Corporate Context

The National University of Sciences and Technology (NUST) Chip Design Centre (NCDC), located in Islamabad, Pakistan, is a premier research and development institution dedicated to advancing silicon design, embedded systems, and microelectronics. NCDC operates at the intersection of academic research and commercial foundry tape-outs, providing highly specialized engineering services in digital, analog, and mixed-signal IC design. The facility hosts the NUST Internship Programme for International Students (NIPIS), fostering global engineering collaboration by integrating international students into active, high-stakes silicon layout workflows targeting nodes such as TSMC 65nm [4].

2 Department Oversight

The internship was hosted within NCDC's Digital ASIC Backend Division under the direct supervision of Dr. Hammad Mehmood Cheema, alongside technical mentorship from senior design engineers Junaid, Fawad, and Hira. This specific unit is responsible for translating verified front-end SystemVerilog and Verilog RTL into manufacturable physical geometries [4]. The division's operational domain encompasses logic synthesis in Cadence Genus, formal verification in Cadence Conformal, floorplanning and routing in Cadence Innovus, and rigorous physical verification (DRC/LVS) using Cadence Tempus and PVS [4].

3 Task Matrices

The training progressed through a highly granular, milestone-driven technical workflow to build proficiency in the Cadence digital flow. To accurately document this progression, the tasks are divided into front-end logic, foundational physical training, and the final processor capstone.

3.1 RTL Datapath & Control Architecture

Authored the SystemVerilog behavioral RTL for a 32-bit RV32I RISC-V processor [3]. Resolved synchronous reset vulnerabilities in the register file by strictly enforcing non-blocking assignments within the `always_ff` block [3]. Expanded the ALU datapath to

support shift logical, shift arithmetic, and set-less-than operations, utilizing `$signed()` casting to ensure arithmetic accuracy during right-shifts [3]. To prevent datapath collisions during branch evaluations, a dedicated branch comparator was implemented to isolate Zero, Negative, Carry, and Overflow flag generation from the main ALU execution pipeline [3]. The design was fully validated via an assembly sequence in Vivado, confirming stable masked memory updates and control flow branching [3].

3.2 Synthesis & Foundational Physical Constraints

To baseline TSMC 65nm cell performance, initial physical-aware synthesis was executed on a 4-bit counter using Cadence Genus. Pushing the target clock constraint from a relaxed 10.0 ns down to an aggressive 1.0 ns forced the synthesis engine to upsize standard cells to high-drive-strength `SDFCNQD4` flip-flops [4]. This resulted in a dynamic power spike from 5.82 μW to 74.88 μW [4]. Logic Equivalence Checking (LEC) via Cadence Conformal reported a primitive count increase from 8 to 12 library cells; this discrepancy was successfully identified as the physical insertion of `CKBD4` clock buffers during CTS, confirming 100% mathematical boolean equivalence remained intact [4].

For integration training, an 8-bit ALU core was bounded with a physical I/O pad ring using TSMC 65nm `PDDW16DGZ` bidirectional pads and `PCORNER` cells [4]. The layout was constrained using strict `set_input_delay` and `set_output_delay` SDC rules to accurately model external pin capacitance prior to routing [4].

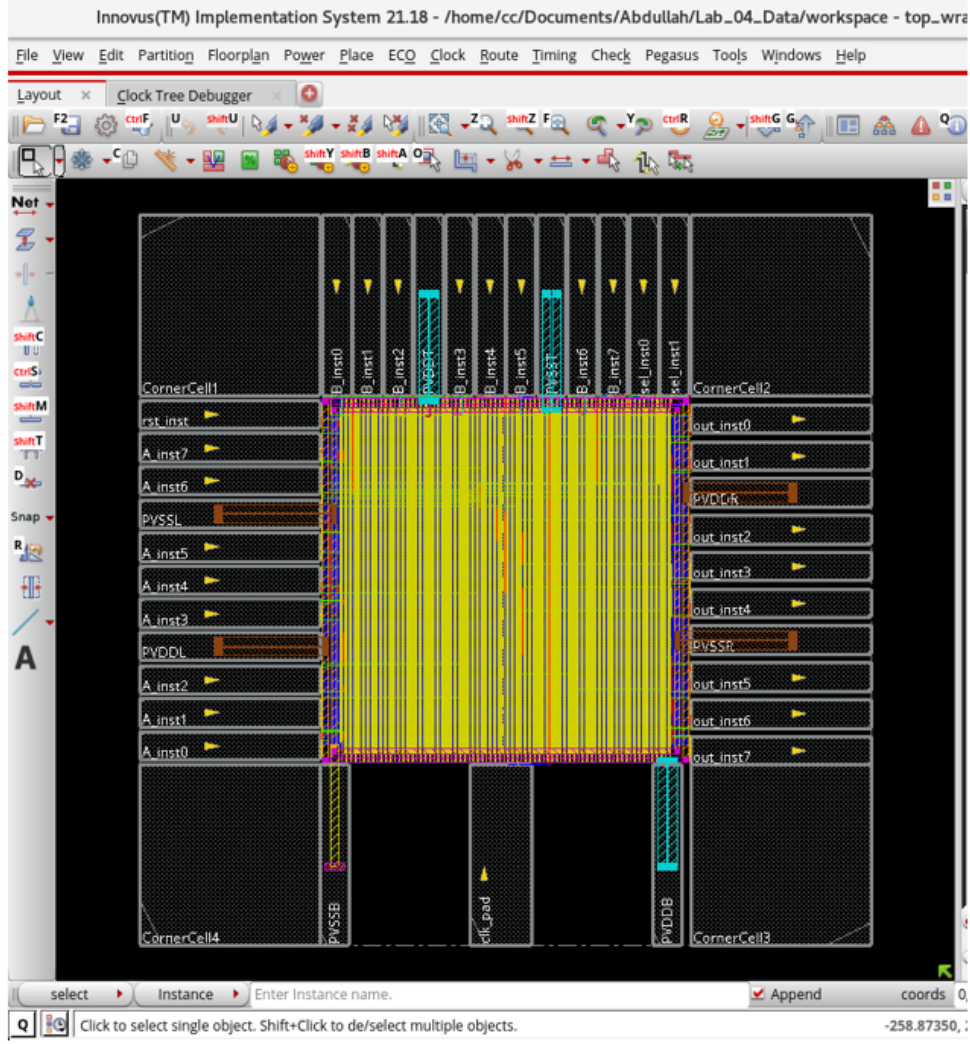


Figure 1: 8-bit ALU Physical Design: Final layout incorporating TSMC 65nm bidirectional pads (PDDW16DGZ) and core filler cells.

3.3 RISC-V Capstone: Floorplanning & Power Mesh

Engineered a $350.0 \mu\text{m} \times 350.0 \mu\text{m}$ core macro for the RISC-V processor, utilizing a custom TCL script to evenly distribute 168 boundary I/O ports across the M3 and M4 metal layers [4]. Constructed a highly dense Power Distribution Network (PDN) utilizing M6/M7 global rings and 152 M4 vertical stripes [4]. The router dropped over 47,000 multi-layer vias down to the M1 follow-pins to prevent dynamic IR drop across the grid [4].

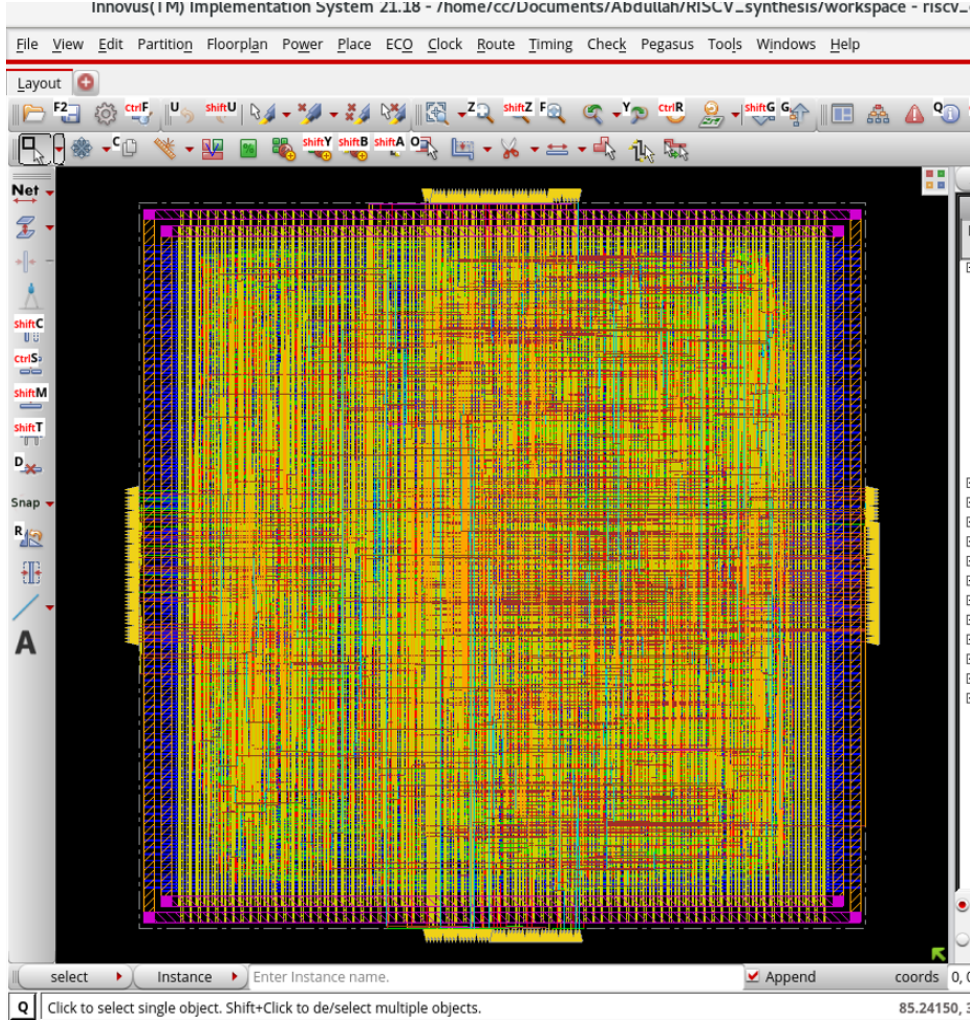


Figure 2: Cadence Innovus: Fully routed RISC-V core showing M4 vertical power stripes and dense standard cell placement.

3.4 Routing, DFM, & Physical Signoff

Achieved 100% routing completion across 10,983 nets using the NanoRoute engine. To maximize manufacturing yield and prevent open-circuit defects, 58,839 single-cut vias were upgraded to multi-cut configurations via `route_design -via_opt`, achieving a 74.6% redundancy conversion rate [4].

Conducted multi-corner static timing analysis in Cadence Tempus utilizing post-route SPEF parasitics and On-Chip Variation (OCV) modeling [4]. Secured a clock frequency of 174.52 MHz (5.73 ns period) with a total power consumption of 5.41 mW [4]. The internal Reg2Reg setup achieved a positive Worst Negative Slack (WNS) of +1.700 ns alongside zero DRC/LVS physical violations [4].

```
#####
# Generated by:      Cadence Innovus 21.18-s099_1
# OS:               Linux x86_64 (Host ID ncdc-0137)
# Generated on:      Wed Aug 12 09:05:35 2026
# Design:           riscv_core
# Command:          check_drc -out_file /home/cc/Documents/Abdullah/RISCV_synthesis/export/Innovus_Data
#####
No DRC violations were found
```

Figure 3: Cadence Innovus Terminal: Final physical verification confirming 0 DRC violations for the RISC-V core prior to GDSII stream-out.

4 Curriculum Connection

The physical design workflow heavily leveraged digital logic fundamentals covered in the Communications and Information Engineering curriculum at Zewail City. Architectural principles concerning dynamic execution, arithmetic pipelining, and data hazards were directly applied during the RTL validation phase [3]. Furthermore, understanding setup and hold margins was critical for optimizing timing loops within the Tempus MMMC environment [4].

Proposed Curriculum Addition: While the current Zewail City curriculum provides exceptional front-end FPGA synthesis training, there is a distinct gap in backend ASIC implementation. Integrating Cadence Innovus into senior-level VLSI coursework—specifically focusing on Multi-Mode Multi-Corner (MMMC) setup scripting and standard cell library extraction—would dramatically optimize student readiness for global semiconductor roles.

5 Practical Innovation Contribution

Workflow Process Suggestion for NCDC:

During the internship, transitioning from Cadence Genus synthesis to Cadence Innovus physical design required the manual mapping of `.lib`, `.lef`, and `.captable` files across multiple PVT corners [4]. I formally recommend the development of an automated MMMC TCL generation script for the lab environment. Automating this environment alignment will eradicate human setup errors, accelerate the synthesis-to-layout handoff, and streamline future silicon tape-outs for incoming engineering cohorts.

Senior Design Capstone Integration:

The exposure to TSMC 65nm standard cell integration directly informs my upcoming senior graduation project. I plan to apply these precise physical optimization and DFM techniques toward designing an on-chip hardware accelerator deployed on an FPGA or SoC. This architecture will be tailored for heavy AI applications, such as real-time surveillance analytics or industrial defect detection.

6 Formal Recognitions

- **NIPIS '26 Internship Certificate:** Officially awarded by NCDC leadership upon the successful completion of the RTL-to-GDSII TSMC 65nm layout and final oral defense.

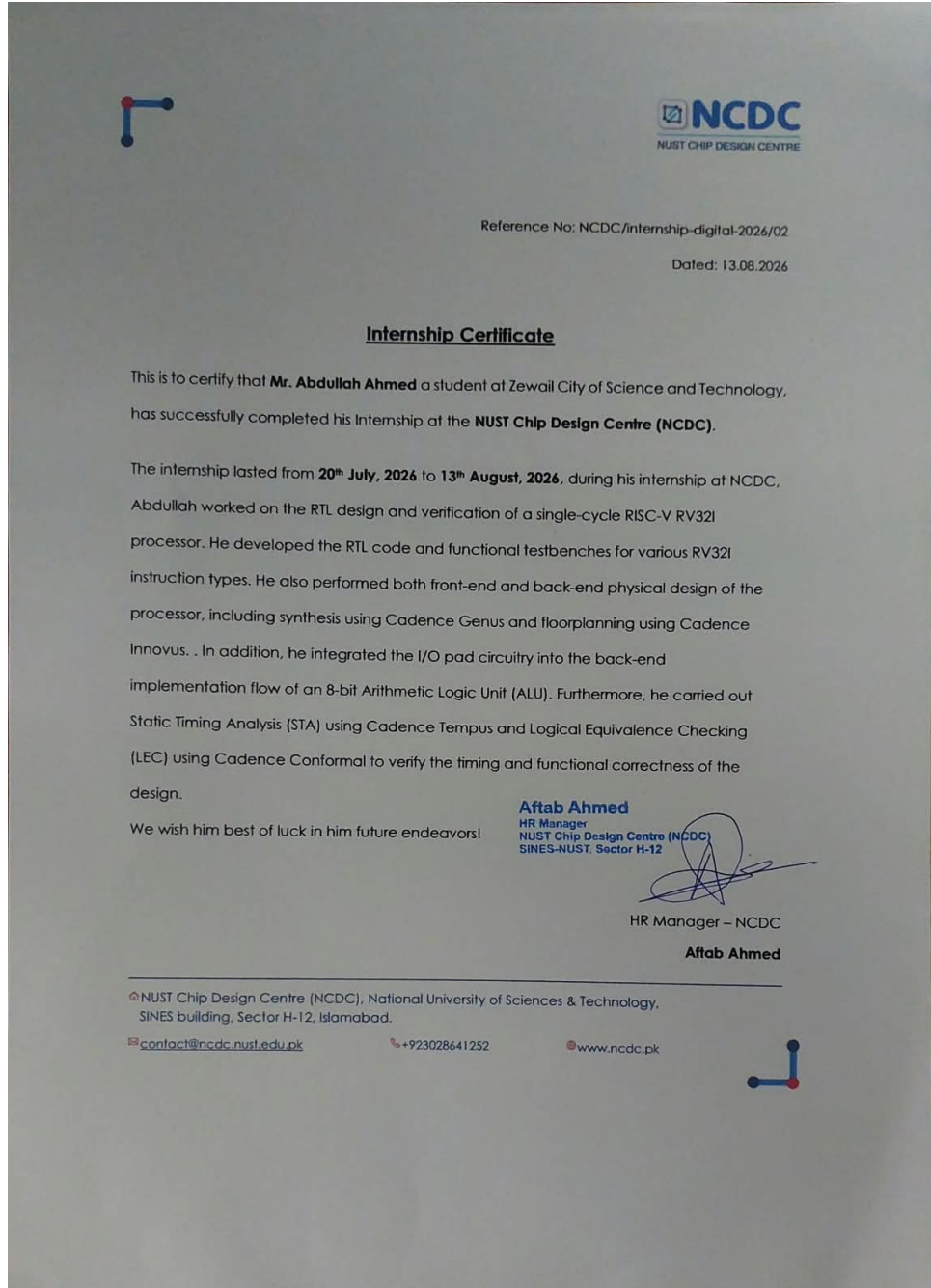


Figure 4: NIPIS '26 Completion: Official awarding of the ASIC Physical Design internship certificate by Dr. Hammad Mehmood Cheema at NCDC.

7 Concluding Remarks

The NIPIS '26 internship at NCDC facilitated a critical engineering transition from abstract software-based SystemVerilog design to managing the tangible physics of actual silicon manufacturing [3, 4]. Overcoming timing degradation, resolving routing congestion on a RISC-V architecture, and manually closing setup and hold violations have provided an invaluable, industry-standard foundation in full-stack ASIC physical design.

8 References & Sourcing Integrity

Note on Corporate Source Confidentiality: Because this project utilizes proprietary Foundry Process Design Kits (PDKs) from TSMC and restricted EDA tools from Cadence Design Systems, linking to public URLs violates strict corporate Non-Disclosure Agreements (NDAs). Consequently, all citations refer to internal engineering logs, laboratory reports, and authorized restricted documentation. The highly specific technical metrics and embedded tool screenshots within this report serve as the primary validating artifacts.

References

- [1] NUST Chip Design Centre (NCDC) Documentation and TSMC 65nm LP Node Design Rule Manuals.
- [2] Cadence Design Systems: Innovus Implementation System, Genus Synthesis Solution, and Tempus Timing Signoff User Guides.
- [3] RISC-V Datapath Implementation Progress Logs.
- [4] Physical Design Technical Analysis Reports (Lab 1-4, RISC-V Capstone).